

Random Variables Overview

- Probability
- Random variables
- Transforms of PDFs
- Moments and cumulants
- Useful distributions
- Random vectors
- Linear transformations of random vectors
- The multivariate normal distribution
- Sums of independent random variables
- Central limit theorem

Taylor Series Expansion

Taylor series expansion about $t = a$,

$$x(t) = x(a) + \frac{dx(t)}{dt}(t-a) + \frac{d^2x(t)}{d^2t} \frac{(t-a)^2}{2!} + \cdots + \frac{d^n x(t)}{d^n t} \frac{(t-a)^n}{n!} + r_n$$

where

$$r_n = \left. \frac{d^{n+1}x(t)}{dt} \right|_{t=\tau \in [t,a]} (t-a)^{n+1}$$

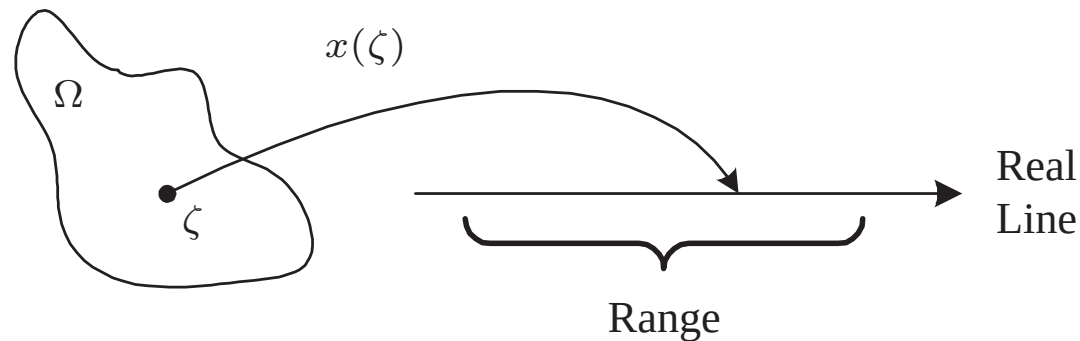
Introduction

- Random signals evolve in time in an unpredictable manner
- We must assume something doesn't change in order to use them
- Usually this is their average properties

Probability Space

- Let Ω denote all possible outcomes, ζ , of an experiment
- **Event**: A subset of outcomes
- The event is said to occur if the outcome of the experiment is one of the members of the subset
 - It's an “or” (union)
 - Not an “and” (intersection)
- A collection of subsets with certain properties is called a **field** and will be denoted as $\mathcal{F}$
- The probability of each event in the field is denoted $\Pr\{\cdot\}$ for $k = 1, 2, \dots$
- The collection $(\Omega, \mathcal{F}, \Pr\{\cdot\})$ is called a **probability space**

Random Variables



- **Random variable:** a function that assigns a real number, $x(\zeta)$, to each outcome ζ in the sample space of a random experiment
- Conditions
 - $\{\zeta : \zeta \in \Omega \text{ and } -\infty < x(\zeta) \leq x_0\} \in \mathcal{F} \text{ for all } x_0 \in \mathbb{R}^1$
 - $\Pr \{x(\zeta) = \infty\} = 0$
 - $\Pr \{x(\zeta) = -\infty\} = 0$

Random Variable Properties

- The sample space Ω is the **domain** of the random variable
- The set of all values that $x(\cdot)$ can have is the **range** of the random variable
- For now the range is the real line, $\mathbb{R}^1$
- Later we will generalize to allow it to be a vector or a sequence
- This is a many to one mapping. That is, a set of points, $\zeta_1, \zeta_2, \dots$ may take on the same value of the random variable
- Will sometimes abbreviate RV
- The RV may be real- or complex-valued
- Random variables are actually a *deterministic function* of any event in the field: $\{\zeta_1, \zeta_2, \dots, \zeta_k\}$

Random Variable Properties Continued

- Notation: $x(\zeta) = x$
 - $x(\zeta)$ denotes a random variable (function of the experimental outcome)
 - x denotes its value
- A random variable may be **Discrete-Valued**, **Continuous-Valued**, or **Mixed**
- Similar to spectral densities
 - If $x(t)$ is nearly periodic, it has discrete spectra
 - Most stationary random signals $x(t)$ have continuous spectra
 - A combination of the two types has mixed spectra

Cumulative Distribution Function (CDF)

- Recall that probability is defined on the field of possible events
- Some of these events (subsets of outcomes) can be defined as $\{\zeta : x(\zeta) \leq x\}$
- **Cumulative Distribution Function (CDF):** The probability of these events, $\Pr \{x(\zeta) \leq x\}$
- Note the CDF is a function of $x \in \mathbb{R}^1$:

$$F_x(x) \triangleq \Pr \{x(\zeta) \leq x\}$$

- The CDF is continuous from the right

Properties of the CDF

- $0 \leq F_x(x) \leq 1$
- $\lim_{x \rightarrow +\infty} F_x(x) = 1$
- $\lim_{x \rightarrow -\infty} F_x(x) = 0$
- $F_x(x)$ is a nondecreasing function of x
 - Thus, if $a < b$, then $F_x(a) \leq F_x(b)$
- $F_x(x)$ is continuous from the right
 - That is, for $h > 0$, $F_x(b) = \lim_{h \rightarrow 0} F_x(b + h) = F_x(b^+)$
- $\Pr \{a < x(\zeta) \leq b\} = F_x(b) - F_x(a)$
- $\Pr \{x(\zeta) = b\} = F_x(b) - F_x(b^-)$

Probability Density Function (PDF)

- **Probability Density Function (PDF):** When it exists, is defined as

$$f_x(x) \triangleq \frac{dF_x(x)}{dx}$$

- Thus, we also have

$$F_x(x) = \int_{-\infty}^x f_x(u) du$$

- For a small interval Δx , can be thought of as

$$f_x(x)\Delta x \approx F_x(x + \Delta x) - F_x(x) = \Pr \{x < x(\zeta) \leq x + \Delta x\}$$

Properties of the PDF

- $f_x(x) \geq 0$
- $\Pr \{a \leq x(\zeta) \leq b\} = \int_a^b f_x(u) \, du$
- $F_x(x) = \int_{-\infty}^x f_x(u) \, du$
- $\int_{-\infty}^{+\infty} f_x(u) \, du = 1$
- A valid PDF can be formed from any nonnegative, piecewise continuous function $g(x)$ that has a finite integral
- The PDF must be defined for all real values of x
- If x does not take on some values, this implies $f_x(x) = 0$ for those values

Point Statistics, Averages, and Moments

- In general, we will often not have a complete description of an RV (the PDF or CDF)
- Estimating a PDF or CDF is difficult in general, especially if $x(\zeta)$ is a sequence or vector
- However, we can often estimate some properties of the distribution without estimating the distribution itself
- These are called **point statistics**
- We will only discuss a subset of point statistics: **statistical averages** or **moments**
- These are useful because
 - In many cases, we can estimate them accurately from data
 - They give us useful information about the distribution
 - We don't have to know the distribution to estimate them

Expected Values Defined

Expected value: defined for a random variable $x(\zeta)$ as

$$E[x(\zeta)] \triangleq \mu_x = \int_{-\infty}^{+\infty} x f_x(x) dx$$

If $x(\zeta)$ is discrete-valued, the PDF consists only of impulses and can be alternatively written in terms of the pmf

$$\begin{aligned} E[x(\zeta)] &= \int_{-\infty}^{+\infty} x \left(\sum_k p_k \delta(x - x_k) \right) dx \\ &= \sum_k p_k \int_{-\infty}^{+\infty} x \delta(x - x_k) dx \\ &= \sum_k p_k x_k \end{aligned}$$

Expectation Properties

$$\mathbb{E}[x(\zeta)] \triangleq \mu_x = \int_{-\infty}^{+\infty} x f_x(x) \, dx$$

- Also called the **mean** of $x(\zeta)$
- The mean can be regarded as the “center of gravity” of $f_x(x)$
- If $f_x(x)$ is symmetric about a , $f(x - a) = f(a - x)$, then $\mu_x = a$
- If $f_x(x)$ is even, $\mu_x = 0$
- This is a linear operation

$$\mathbb{E}[\alpha x(\zeta) + \beta] = \alpha \mu_x + \beta$$

- If $y(\zeta) = g(x(\zeta))$, a function of the random variable $x(\zeta)$, $y(\zeta)$ is also a random variable such that

$$\mathbb{E}[y(\zeta)] \triangleq \mathbb{E}[g(x(\zeta))] = \int_{-\infty}^{\infty} g(x) f_x(x) \, dx$$

Moments Defined

m th-order Moment of $x(\zeta)$ is defined as

$$r_x^{(m)} \triangleq \text{E} [x^m(\zeta)] = \int_{-\infty}^{\infty} x^m f_x(x) \, dx$$

m th-order Central Moment of $x(\zeta)$ is defined as

$$\gamma_x^{(m)} \triangleq \text{E} [(x(\zeta) - \mu_x)^m] = \int_{-\infty}^{\infty} (x - \mu_x)^m f_x(x) \, dx$$

- **Mean-Squared Value:** The second-order moment, $r_x^{(2)}$.
- Note that

$$\text{E} [x^2(\zeta)] \neq \text{E}^2 [x(\zeta)]$$

Moments and Definitions

$$r_x^{(m)} \triangleq \mathbb{E}[x^m(\zeta)] = \int_{-\infty}^{\infty} x^m f_x(x) dx$$

$$\gamma_x^{(m)} \triangleq \mathbb{E}[(x(\zeta) - \mu_x)^m] = \int_{-\infty}^{\infty} (x - \mu_x)^m f_x(x) dx$$

- **Variance** of $x(\zeta)$ is defined as

$$\text{var}[x(\zeta)] \triangleq \sigma_x^2 \triangleq \gamma_x^{(2)} = \mathbb{E}[(x(\zeta) - \mu_x)^2]$$

- **Standard deviation** of $x(\zeta)$ is defined as $\sigma_x = \sqrt{\text{var}[x(\zeta)]}$
- Obvious moments

$$r_x^{(0)} = 1$$

$$r_x^{(1)} = \mu_x$$

- Trivial central moments

$$\gamma_x^{(0)} = 1$$

$$\gamma_x^{(1)} = 0$$

$$\gamma_x^{(2)} = \sigma_x^2$$

Relationship of Moments

Moments and central moments are related:

$$\gamma_x^{(m)} = \sum_{k=0}^m \binom{m}{k} (-1)^k \mu_x^k r_x^{(m-k)}$$

It also possible to calculate the first m moments from the first m central moments and the mean μ_x .

Characteristic Functions

Characteristic Function of $x(\zeta)$ is defined as

$$\Phi(\xi) \triangleq \mathbb{E} \left[e^{j\xi x(\zeta)} \right] = \int_{-\infty}^{\infty} f_x(x) e^{j\xi x} dx$$

- Similar to the continuous-time Fourier transform
- However, the exponent is positive (why?)
- Will not use $F_x(\xi)$ to avoid confusion with CDF (same as text)
- Text claims the independent variable ξ should not be thought of as frequency (why?)
- This will be useful for handling sums of random variables because the distribution of the sum is the convolution of the PDF's

Moment Generating Functions

Moment Generating Function of $x(\zeta)$ is defined as

$$\bar{\Phi}(s) \triangleq \text{E} \left[e^{sx(\zeta)} \right] = \int_{-\infty}^{\infty} f_x(x) e^{sx} dx$$

- Similar to the continuous-time Laplace transform
- However, the exponent is positive (why?)

Using a Taylor series expansion of e^{sx} at x close to zero we have

$$\begin{aligned} \bar{\Phi}(s) &= \text{E} \left[e^{sx(\zeta)} \right] \\ &= \text{E} \left[1 + sx(\zeta) + \frac{[sx(\zeta)]^2}{2!} + \dots + \frac{[sx(\zeta)]^m}{m!} + \dots \right] \\ &= 1 + s\mu_x + \frac{s^2}{2!} r_x^{(2)} + \dots + \frac{s^m}{m!} r_x^{(m)} + \dots \end{aligned}$$

Moment Generating Functions

$$\bar{\Phi}(s) = 1 + s\mu_x + \frac{s^2}{2!}r_x^{(2)} + \cdots + \frac{s^m}{m!}r_x^{(m)} + \cdots$$

- If all the moments of $x(\zeta)$ are known and exist, we can create $\bar{\Phi}(s)$ and solve for $f_x(x)$ by inverse Laplace transform
- Thus the set of all moments (if they exist) completely define the PDF!
- If $\bar{\Phi}(s)$ is known and analytic (not in the book), we can use it to solve for the moments

$$r_x^{(m)} = \left. \frac{d^m[\bar{\Phi}(s)]}{ds^m} \right|_{s=0} = (-j)^m \left. \frac{d^m[\bar{\Phi}(\xi)]}{d\xi^m} \right|_{\xi=0}$$

Cumulants

Cumulant Generating Function of $x(\zeta)$ is defined as

$$\bar{\Psi}_x(s) \triangleq \ln \bar{\Phi}_x(s) = \ln E \left[e^{sx(\zeta)} \right]$$

Second Characteristic Function of $x(\zeta)$ is defined as

$$\Psi_x(\xi) \triangleq \ln \Phi_x(\xi) = \ln E \left[e^{j\xi x(\zeta)} \right]$$

m th Cumulant of $x(\zeta)$ is defined as

$$\kappa_x^{(m)} \triangleq \left. \frac{d^m [\bar{\Psi}(s)]}{ds^m} \right|_{s=0} = (-j)^m \left. \frac{d^m [\bar{\Psi}(\xi)]}{d\xi^m} \right|_{\xi=0}$$

Useful for high-order moment analysis (advanced SSP topic) and working with products of characteristic functions

Common Random Variables

- Uniform distribution
- Normal distribution
- Cauchy distribution

Uniform Distribution

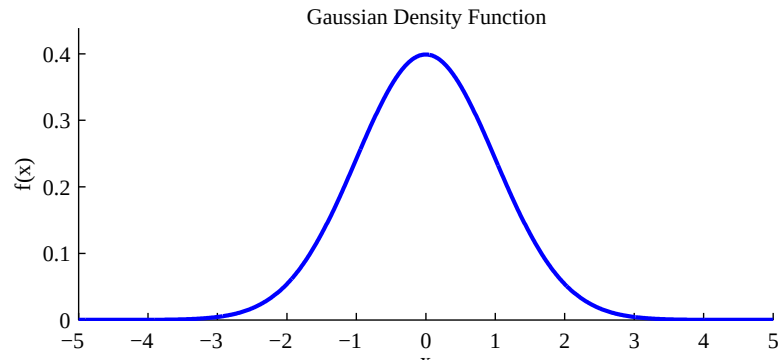
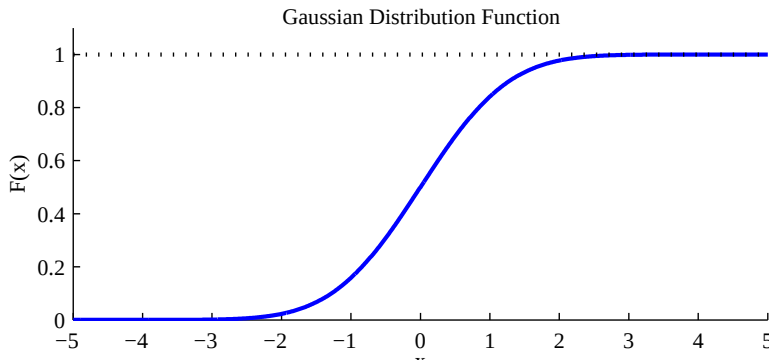
$$f_x(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

- Useful for situations in which outcomes are equally likely
- Often denoted as $x(\zeta) \sim \mathcal{U}[a, b]$
- Mean and variance

$$\mu_x = \frac{a + b}{2}$$

$$\sigma_x^2 = \frac{(b - a)^2}{12}$$

Normal Distribution



$$f_x(x) = \frac{1}{\sqrt{2\pi\sigma_x^2}} \exp\left(-\frac{(x - \mu_x)^2}{2\sigma_x^2}\right)$$

- Also called the **Gaussian** distribution
- Often denoted as $x(\zeta) \sim \mathcal{N}(\mu_x, \sigma_x^2)$
- Arises naturally in many applications
- Central limit theorem (more later)

Normal Distribution Properties

$$f_x(x) = \frac{1}{\sqrt{2\pi\sigma_x^2}} e^{-\frac{(x-\mu_x)^2}{2\sigma_x^2}}$$
$$\Phi_x(\xi) = \exp(j\mu_x\xi - \frac{1}{2}\sigma_x^2\xi^2)$$

- Defined completely by μ_x and σ_x^2
- All higher-order moments can be determined in terms of the first two moments
- Higher-order moments provide no additional information
- Due to the central limit theorem, would like to know how the distribution of random variables differs from a normal distribution
- Cumulants generally provide this information
- For normally distributed random variables, $\kappa_x^{(m)} = 0$ for $m > 2$

Cauchy Random Variable

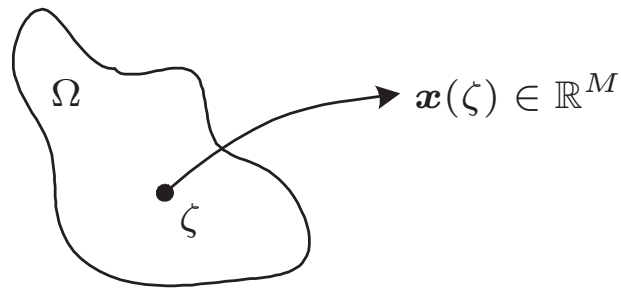
$$f_x(x) = \frac{\beta}{\pi} \frac{1}{(x - \mu)^2 + \beta^2}$$

$$F_x(x) = 0.5 + \frac{1}{\pi} \arctan \frac{x - \mu}{\beta}$$

$$\Phi_x(\xi) = \exp(j\mu\xi - \beta|\xi|)$$

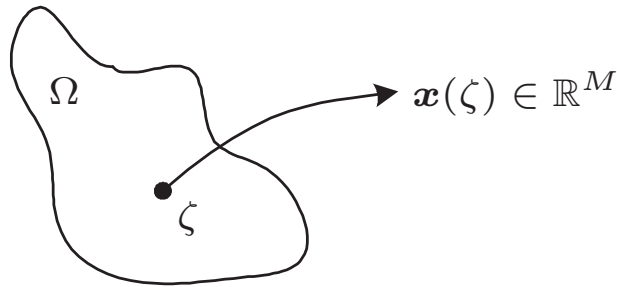
- A heavy-tailed distribution (relative to Gaussian)
- Two parameters, μ and β
- Mean: $\mu_x = \mu$
- Variance does not exist because $E[x^2]$ does not exist
- Moment generating function does not exist for some values of s
- The sum of M independent Cauchy random variables is also Cauchy!
- Example of an infinite-variance random variable

Random Vectors



- Much of what we have discussed generalizes to vector random-variables in an obvious manner
- However, the lower order moments have special properties and are important to signal processing
- Each outcome of the assumed underlying random experiment produces an entire random vector
- Each element of the vector is not generated independently from a separate experiment
- This is an important concept

Random Vectors



Random M vector: a real-valued vector containing M random variables

$$\mathbf{x} = [x_1(\zeta), x_2(\zeta), \dots, x_M(\zeta)]^T$$

- Transpose is denoted by T
- Maps an abstract probability space to a vector-valued real space $\mathbb{R}^M$
- Thus the range is an M -dimensional space

Distribution and Density Functions

$$F_{\mathbf{x}}(x_1, \dots, x_M) \triangleq \Pr \{x_1(\zeta) \leq x_1, \dots, x_M(\zeta) \leq x_M\}$$

$$f_{\mathbf{x}}(\mathbf{x}) = \lim_{\substack{\Delta x_1 \rightarrow 0 \\ \vdots \\ \Delta x_M \rightarrow 0}} \frac{\Pr \{x_1 < x_1(\zeta) \leq x_1 + \Delta x_1, \dots, x_M < x_M(\zeta) \leq x_M + \Delta x_M\}}{\Delta x_1 \dots \Delta x_M}$$

- The commas denote an *and* condition: it is the probability that all M random variables $x_i(\zeta)$ are less than the stated values
- A RV is completely characterized by its joint CDF or PDF
- Often written as $F_{\mathbf{x}}(\mathbf{x}) = \Pr \{\mathbf{x}(\zeta) \leq \mathbf{x}\}$

Distribution and Density Functions

$$\begin{aligned}f_{\mathbf{x}}(\mathbf{x}) &\triangleq \frac{\partial}{\partial x_1} \cdots \frac{\partial}{\partial x_m} F_{\mathbf{x}}(\mathbf{x}) \\F_{\mathbf{x}}(\mathbf{x}) &= \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_M} f_{\mathbf{x}}(\mathbf{u}) \, du_1 \cdots du_M \\&= \int_{-\infty}^{\mathbf{x}} f_{\mathbf{x}}(\mathbf{u}) \, d\mathbf{u}\end{aligned}$$

Independence

Independent Two random variables $x_1(\zeta)$ and $x_2(\zeta)$ are independent if the events $\{x_1(\zeta) \leq x_1\}$ and $\{x_2(\zeta) \leq x_2\}$ are jointly independent

$$\Pr \{x_1(\zeta) \leq x_1, x_2(\zeta) \leq x_2\} = \Pr \{x_1(\zeta) \leq x_1\} \Pr \{x_2(\zeta) \leq x_2\}$$

Equivalent conditions are

$$\begin{aligned} F_{x_1, x_2}(x_1, x_2) &= F_{x_1}(x_1)F_{x_2}(x_2) \\ f_{x_1, x_2}(x_1, x_2) &= f_{x_1}(x_1)f_{x_2}(x_2) \end{aligned}$$

Vector Statistics

- In general it is very difficult to estimate vector PDF's and/or CDF's
- Called the “curse of dimensionality”
- Even if they are known, they are difficult to work with in general
- However, there is a rich statistical theory developed for second order moments (mean and variance)
- We will focus on this aspect of SSP for this course
- Higher-order moments is an advanced topic covered later in the text
 - Worth self-study

Vector Mean

Mean Vector: defined for a random variable $\mathbf{x}(\zeta)$ as

$$\boldsymbol{\mu}_{\mathbf{x}} = \mathbb{E} [\mathbf{x}(\zeta)] = \begin{bmatrix} \mathbb{E} [x_1(\zeta)] \\ \vdots \\ \mathbb{E} [x_M(\zeta)] \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_M \end{bmatrix}$$

- The integral of the vector expectation is taken over the entire $\mathbb{C}^M$ space, if the random vector is complex-valued
- The mean vector is simply the vector of means

Autocorrelation Matrix

Autocorrelation matrix: defined for a random vector $\mathbf{x}(\zeta)$ as

$$\mathbf{R}_x \triangleq \text{E} [\mathbf{x}(\zeta) \mathbf{x}^H(\zeta)] = \begin{bmatrix} r_{11} & \dots & r_{1M} \\ \vdots & \ddots & \vdots \\ r_{M1} & \dots & r_{mm} \end{bmatrix}$$

where

$$r_{ii} \triangleq \text{E} [|x_i(\zeta)|^2] \qquad r_{ij} \triangleq \text{E} [x_i(\zeta) x_j(\zeta)^*] = r_{ji}^*$$

- H denotes conjugate transpose operation
- This is a completely different definition than the autocorrelation defined for deterministic signals
- r_{ii} are second-order moments of $x_i(\zeta)$, denoted earlier as $r_{x_i}^{(2)}$
- r_{ij} measure *correlation* between two random variables
- This will be precisely defined later
- The autocorrelation matrix $\mathbf{R}_x$ is Hermitian: $\mathbf{R}_x = \mathbf{R}_x^H$

Autocovariance Matrix

Autocovariance matrix: defined for a random variable $\mathbf{x}(\zeta)$ as

$$\mathbf{\Gamma}_{\mathbf{x}} \triangleq \text{E} [(\mathbf{x}(\zeta) - \boldsymbol{\mu}_{\mathbf{x}})(\mathbf{x}(\zeta) - \boldsymbol{\mu}_{\mathbf{x}})^{\text{H}}] = \begin{bmatrix} \gamma_{11} & \cdots & \gamma_{1M} \\ \vdots & \ddots & \vdots \\ \gamma_{M1} & \cdots & \gamma_{MM} \end{bmatrix}$$

where

$$\gamma_{ii} \triangleq \text{E} [|x_i(\zeta) - \mu_i|^2] = \sigma_{x_i}^2$$

$$\gamma_{ij} \triangleq \text{E} [(x_i(\zeta) - \mu_i)(x_j(\zeta) - \mu_j)^*] = \text{E} [x_i(\zeta)x_j^*(\zeta)] - \mu_i\mu_j = \gamma_{ji}^*$$

- Sometimes γ_{ij} is denoted as σ_{ij}
- γ_{ii} is sometimes called the **self variance** of $x_i(\zeta)$
- γ_{ij} is called the **covariance** of $x_i(\zeta)$ and $x_j(\zeta)$
- The covariance matrix $\mathbf{\Gamma}_{\mathbf{x}}$ is Hermitian: $\mathbf{\Gamma}_{\mathbf{x}} = \mathbf{\Gamma}_{\mathbf{x}}^{\text{H}}$

Correlation Coefficients

Correlation coefficient: is denoted as ρ_{ij} and is defined for random variables $x_i(\zeta)$ and $x_j(\zeta)$ as

$$\gamma_{ij} = \rho_{ij}\sigma_i\sigma_j \qquad \rho_{ij} \triangleq \frac{\gamma_{ij}}{\sigma_i\sigma_j}$$

- $\rho_{ii} = 1$ and $-1 \leq \rho_{ij} \leq 1$
- Extreme values of ρ_{ij} indicate a linear relationship between $x_j(\zeta)$ and $x_i(\zeta)$: $x_j(\zeta) = \alpha x_i(\zeta) + \beta$
 - Does not tell you what α or β are
 - $\rho_{ij} = 1$ implies $\alpha > 0$, $\rho_{ij} = -1$ implies $\alpha < 0$
- If $\rho_{ij} = 0$, $x_i(\zeta)$ and $x_j(\zeta)$ are said to be **uncorrelated**

Autocorrelation and Autocovariance

$$\Gamma_x = R_x - \mu_x \mu_x^H$$

- If $\mu_x = 0$, $\Gamma_x = R_x$
- If $\rho_{ij} = 0$ or $\gamma_{ij} = 0$, $x_i(\zeta)$ and $x_j(\zeta)$ are **uncorrelated**
- If $r_{ij} = 0$, $x_i(\zeta)$ and $x_j(\zeta)$ are **orthogonal**
- The degree of correlation is a weaker measure of interaction than independence
 - If $x_i(\zeta)$ and $x_j(\zeta)$ are independent, $\gamma_{ij} = \rho_{ij} = 0$
 - Converse is not true (sufficient, but not a necessary condition)
 - If $x_i(\zeta)$ and $x_j(\zeta)$ have a normal distribution, then $\gamma_{ij} = \rho_{ij} = 0$ implies $x_i(\zeta)$ and $x_j(\zeta)$ are independent
- If one or both random variables have zero mean and are uncorrelated, they are orthogonal

Cross-Correlation and Cross-Covariance

Cross-correlation matrix: defined for a random vectors $\mathbf{x}(\zeta) \in \mathbb{C}^M$ and $\mathbf{y}(\zeta) \in \mathbb{C}^L$ as

$$\mathbf{R}_{xy} \triangleq \mathbb{E} [\mathbf{x}(\zeta) \mathbf{y}^H(\zeta)] = \begin{bmatrix} \mathbb{E} [x_1(\zeta) y_1^*(\zeta)] & \dots & \mathbb{E} [x_1(\zeta) y_L^*(\zeta)] \\ \vdots & \ddots & \vdots \\ \mathbb{E} [x_M(\zeta) y_1^*(\zeta)] & \dots & \mathbb{E} [x_M(\zeta) y_L^*(\zeta)] \end{bmatrix}$$

Cross-covariance matrix: defined for a random vectors $\mathbf{x}(\zeta) \in \mathbb{C}^M$ and $\mathbf{y}(\zeta) \in \mathbb{C}^L$ as

$$\mathbf{\Gamma}_{xy} \triangleq \mathbb{E} [(\mathbf{x}(\zeta) - \boldsymbol{\mu}_x)(\mathbf{y}(\zeta) - \boldsymbol{\mu}_y)^H] = \mathbf{R}_{xy} - \boldsymbol{\mu}_x \boldsymbol{\mu}_y^H$$

- **Uncorrelated** if $\mathbf{\Gamma}_{xy} = \mathbf{0}$ or, equivalently, $\mathbf{R}_{xy} = \boldsymbol{\mu}_x \boldsymbol{\mu}_y^H$
- **Orthogonal** if $\mathbf{R}_{xy} = \mathbf{0}$

Linear Transformations

$$\mathbf{y}(\zeta) = \mathbf{A}\mathbf{x}(\zeta)$$

$$\boldsymbol{\mu}_y = \text{E} [\mathbf{y}(\zeta)] = \text{E} [\mathbf{A}\mathbf{x}(\zeta)] = \mathbf{A} \text{E} [\mathbf{x}(\zeta)] = \mathbf{A}\boldsymbol{\mu}_x$$

$$\mathbf{R}_y = \text{E} [\mathbf{y}(\zeta)\mathbf{y}(\zeta)^{\text{H}}] = \text{E} [\mathbf{A}\mathbf{x}(\zeta)\mathbf{x}(\zeta)^{\text{H}}\mathbf{A}^{\text{H}}] = \mathbf{A}\mathbf{R}_x\mathbf{A}^{\text{H}}$$

$$\boldsymbol{\Gamma}_y = \mathbf{A}\boldsymbol{\Gamma}_x\mathbf{A}^{\text{H}}$$

$$\mathbf{R}_{xy} = \mathbf{R}_x\mathbf{A}^{\text{H}}$$

$$\mathbf{R}_{yx} = \mathbf{A}\mathbf{R}_x$$

$$\boldsymbol{\Gamma}_{xy} = \boldsymbol{\Gamma}_x\mathbf{A}^{\text{H}}$$

$$\boldsymbol{\Gamma}_{yx} = \mathbf{A}\boldsymbol{\Gamma}_x$$

Normal Random Vectors

$$f_{\mathbf{x}}(\mathbf{x}) = \frac{1}{(2\pi)^{M/2} |\mathbf{\Gamma}_{\mathbf{x}}|^{\frac{1}{2}}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_{\mathbf{x}})^{\text{H}} \mathbf{\Gamma}_{\mathbf{x}}^{-1} (\mathbf{x} - \boldsymbol{\mu}_{\mathbf{x}}) \right)$$

$$\Phi_{\mathbf{x}}(\boldsymbol{\xi}) = \exp(j \boldsymbol{\xi}^{\text{T}} \boldsymbol{\mu}_{\mathbf{x}} - \frac{1}{2} \boldsymbol{\xi}^{\text{T}} \mathbf{\Gamma}_{\mathbf{x}} \boldsymbol{\xi})$$

- Often denoted as $\mathbf{x}(\zeta) \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{x}}, \mathbf{\Gamma}_{\mathbf{x}})$
- The exponent is a positive definite quadratic function of $\mathbf{x}$
- The PDF is completely specified by $\boldsymbol{\mu}_{\mathbf{x}}$ and $\mathbf{\Gamma}_{\mathbf{x}}$
- Both are relatively easy to estimate in practice
- All higher-order moments can be calculated from these
- If all pairs uncorrelated, are also independent
- A linear transformation, $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{b}$ is also normally distributed, $\mathbf{y}(\zeta) \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{x}} + \mathbf{b}, \mathbf{A}\mathbf{\Gamma}_{\mathbf{x}}\mathbf{A}^{\text{H}})$

Sums of Independent Random Variables

$$y(\zeta) = \sum_{k=1}^M c_k x_k(\zeta) \quad \mu_y = \sum_{k=1}^M c_k \mu_{x_k} \quad \sigma_y^2 = \sum_{k=1}^M |c_k|^2 \sigma_{x_k}^2$$

Let

$$\begin{aligned} y(\zeta) &= x_1(\zeta) + x_2(\zeta) \\ \Phi_y(\xi) &= \mathbb{E} \left[e^{j\xi y(\zeta)} \right] \\ &= \mathbb{E} \left[e^{j\xi [x_1(\zeta) + x_2(\zeta)]} \right] \\ &= \mathbb{E} \left[e^{j\xi x_1(\zeta)} \right] \mathbb{E} \left[e^{j\xi x_2(\zeta)} \right] \\ &= \Phi_{x_1}(\xi) \Phi_{x_2}(\xi) \end{aligned}$$

Therefore

$$f_y(y) = f_{x_1}(y) * f_{x_2}(y)$$

Sums of Independent Random Variables Continued

This generalizes in the obvious manner

$$y(\zeta) = \sum_{k=1}^M x_k(\zeta)$$

$$f_y(y) = f_{x_1}(y) * f_{x_2}(y) * \cdots * f_{x_M}(y)$$

$$\Phi_y(\xi) = \prod_{k=1}^M \Phi_{x_k}(\xi)$$

$$\Psi_y(\xi) = \sum_{k=1}^M \Psi_{x_k}(\xi)$$

$$\kappa_y^{(m)} = \sum_{k=1}^M \kappa_{x_k}^{(m)}$$

Stable Distributions

$$y(\zeta) = \sum_{k=1}^M x_k(\zeta)$$

- **Stable Distributions:** distributions that are preserved (self-reproduce) under convolution
- Examples: Gaussian distribution, Cauchy distribution
- *The only stable distribution with finite variance is the Gaussian distribution*
- The other distributions have infinite variance and possibly infinite mean
- Examples: some diffusion processes, electrical noise, some random walks, coupled oscillators with friction
- Useful to model signals with large variability
- Very good discussion in text (not critical to this class)

Central Limit Theorem

$$y(\zeta) = \sum_{k=1}^M x_k(\zeta)$$

- If $x_i(\zeta)$ are independent identically distributed (IID) random variables and the distribution $f_x(x)$ is stable, clearly $y(\zeta)$ converges to the same distribution as $M \rightarrow \infty$
- **Central Limit Theorem:** If
 - The *mean and variance* of each random variable *exist* (finite)
 - The random variables are *mutually independent*
 - The random variables are *identically distributed*then the distribution of $y(\zeta)$ approaches a normal distribution as $M \rightarrow \infty$

Central Limit Theorem Comments

$$y(\zeta) = \sum_{k=1}^M x_k(\zeta)$$

- The convergence often occurs even if the distributions are not identical
- The convergence is more accurate (rapid) near the mean (center) of the distribution
 - The approximation may be poor in the tails
- The convergence is in the CDF, not the PDF. Consider discrete RV's
- Does not apply when the variance of the RV's is infinite